ADNAN BAIG AI Engineer

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SUMMARY

Software Engineer with 2+ years building Generative AI and Agentic AI systems for enterprise products multi-agent orchestration, LLM tool and MCP integration, and the async Python backends that run them. Architected a multi-agent autonomous testing platform that cut manual testing effort ~80% and human intervention ~85%, and a 9-agent evaluation pipeline that red-teams and validates enterprise AI agents across 20+ quality dimensions. Core stack: Python, FastAPI, LangGraph, LangChain, Google ADK, OpenAI / Gemini / Anthropic, Playwright, PostgreSQL, Snowflake, Databricks, PySpark, Docker.

TECHNICAL SKILLS

Generative AI & LLM: LLM application development, prompt engineering, structured outputs (Instructor), embeddings, semantic search, model fine-tuning, NLP, conversational interfaces

Agentic AI: Multi-agent orchestration, agent memory & state management, tool calling, MCP integrations, Agent-to-Agent (A2A) protocol and agent cards, agent evaluation, LLM red teaming, LangGraph, LangChain, Google ADK, FastAgent

Backend Engineering: Python, FastAPI, REST APIs, async programming, microservices, event-driven systems, Celery, RabbitMQ, Redis, Git, Automations

Data & Databases: PostgreSQL, MySQL, Snowflake, Databricks, Azure SQL, Apache Spark / PySpark, Pandas, NumPy, SQL, ChromaDB, FAISS

ML & Computer Vision: Machine learning, deep learning, Transformers, scikit-learn, TensorFlow, Keras, OpenCV, object detection

Document Processing: PDF parsing (PyMuPDF, pdfplumber, pypdf, pdfminer.six), Tesseract OCR, text-layer detection, AFP-to-PDF conversion, Excel automation

Platform & Tooling: Docker, CI/CD, observability & tracing (Logfire), Playwright, Crawl4AI, Hugging Face, Ollama, NVIDIA AI

PROFESSIONAL EXPERIENCE

Associate Software Engineer (AI/ML) — Zyrix AI **08/2024 – Present**

Hyderabad, India

- Architected and delivered production Agentic AI systems for enterprise clients across autonomous software testing, AI-agent evaluation, and natural-language data analytics owning design, agent architecture, backend services, and deployment.
- Reduced a client's manual testing effort by ~80% and human intervention by ~85% by replacing scripted QA with a multi-agent platform that autonomously designs, validates, and executes test suites.
- Built LLM orchestration layers with agent memory and state management, tool calling, MCP integrations, and structured output enforcement, served through async Python and FastAPI backend services.
- Integrated AI agents with enterprise systems relational databases, cloud data warehouses, Jira, Figma, and REST APIs so agents reason over live business context rather than static prompts.
- Instrumented agent runs with structured logging and tracing to debug non-deterministic multi-agent behaviour and improve execution reliability.

Data Science Intern — Innomatics Research Labs **01/2024 – 04/2024**

Hyderabad, India

- Built and deployed machine learning, NLP, and Generative AI applications for text analysis, automation, and predictive insight, covering the full workflow from data preprocessing and EDA through feature engineering, model training, evaluation, and deployment.

SELECTED AI ENGINEERING PROJECTS

Project deliveries at Zyrix AI

Test Auto Pilot — Agentic AI Autonomous Testing Platform **08/2025 – Present**

Kitap

- Architected a multi-agent platform that autonomously tests web and enterprise applications from any starting artifact a URL, BRD document, Figma design, Jira user story, or plain-language prompt covering public and authenticated web applications plus AI-driven automation of mainframe systems.
- Designed specialised agents for application understanding, requirement analysis, feature discovery, business-flow mapping, manual and BDD test generation, and Playwright-based execution; extended the platform with visual regression testing, accessibility testing, intelligent regression planning, and requirement-compliance analysis.

- Built a self-healing execution service so test runs recover from application and locator drift instead of failing, reducing suite maintenance overhead.
- Engineered the mainframe test automation and document-processing pipeline behind output validation, text-layer detection across multiple PDF parsers, and Tesseract OCR fallback for image-only documents enabling mainframe print output to be validated automatically instead of read by hand.
- Delivered ~80% reduction in manual testing effort and ~85% reduction in human intervention across the full test design, validation, and execution lifecycle.

Tech: Python, Agentic AI, Playwright, Crawl4AI, FastAPI, LLMs, async processing, celery, RabbitMQ, Redis, Docker

Agent2Agent (A2A) AI Evaluation Platform **04/2026 – Present**

ATC

- Built an Agent-to-Agent testing platform that autonomously evaluates third-party AI agents against their declared capabilities using agent cards, multi-agent orchestration, and LLM reasoning providing end-to-end quality assurance for enterprise AI systems.
- Designed a 9-agent evaluation pipeline spanning agent analysis, test planning, multi-turn execution, persona-based testing, security red teaming, compliance validation, regression analysis, and executive reporting, scoring agents across 20+ dimensions including capability adherence, tool usage, safety, robustness, and reliability.

Tech: Python, A2A protocol, multi-agent orchestration, LLM-as-judge evaluation, FastAPI

Data Analytics AI Agents **11/2024 – Present**

DQG

- Built an NLP-powered analytics agent that answers complex business questions in natural language by reasoning jointly over relational database data and requirements documents.
- Implemented anomaly detection and automated reasoning over large-scale enterprise data.
- Engineered a distributed row-wise data comparison system on Apache Spark clusters, optimised for large-dataset reconciliation.
- Unified a single agent interface across five data platforms MySQL, PostgreSQL, Snowflake, Databricks, and Azure SQL.

Tech: Python, LLM orchestration, SQL, Pandas, Agentic AI

Multi-Source Data Deduplication & Entity Resolution Pipeline 08/**2026 – Present**

- Built a deduplication and record-linkage pipeline that consolidates equipment master data from multiple source systems and physical sites into a single trusted record set, replacing manual spreadsheet reconciliation.
- Implemented within-source deduplication with merge-candidate scoring, producing structured Excel deliverables backed by JSON lineage reports that trace every merged record back to its source rows.
- Designed a human-in-the-loop review workbook with per-merge-set approval controls, master-versus-lineage record detail sheets, and a unique-records tab, so data owners could approve or reject merges before they were committed.

Tech: Python, Pandas, SQL, Excel automation, LLM agents, async processing

Computer Vision Safety Monitoring — Face Recognition & PPE/Incident Detection **09/2024 – 11/2024**

- Built a real-time face recognition system trained on large image datasets to detect and identify individuals by name, plus a PPE compliance module monitoring helmets, vests, and gloves in industrial environments.
- Implemented incident detection (including fire) using object detection models, with real-time WhatsApp alerting to notify site administrators of safety breaches.

Tech: Python, OpenCV, deep learning object detection

EDUCATION

B.E., Prof. Ram Meghe Institute of Technology & Research (PRMITR), Badnera — 82.55% 2020 – 2023

Diploma in Engineering, Government College of Polytechnic, Achalpur — 82.26% 2017 – 2020

CERTIFICATIONS

Data Science Professional — Innomatics Research Labs • Machine Learning with Python (powered by IBM) — Innomatics Research Labs